Because Testing Matters

Welcome to the Archimedes Reference Architecture Portal

ARCHIMEDES
RA

Explore comprehensive reference architectures, practical implementations, and interactive demos designed to accelerate your automated **Archimedes** Solution development.

Data Stream

Discover how data streaming powers to fetch all the information related to your testers.

UltraEdge

Explore architectures related to the UltraEdge. This will guide you on the way to communicate with an UltraEdge

Test Program Interactions

Adaptive tests, Rebin, gives you the opportunity to connect to your test program with or without UltraEdge.

Use Cases

Browse practical implementations and scenarios showing how Archimedes accelerates how to unleash the power of Archimedes.

Supplementary

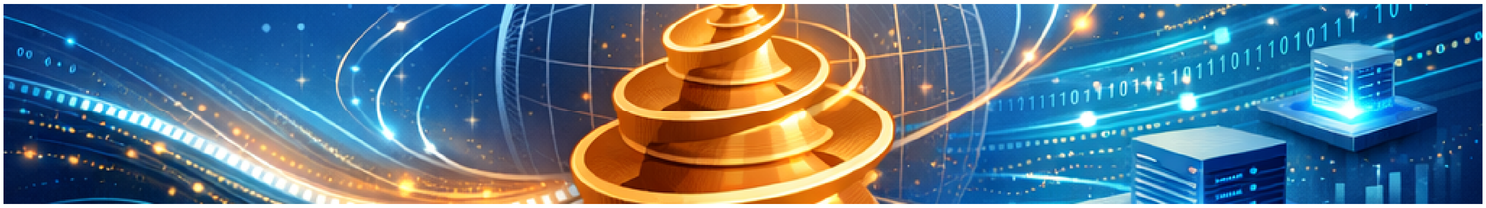
Additional topics, tips, and resources to enhance your Archimedes experience.

Artificial Intelligence

Explore AI-powered features and machine learning integrations for intelligent test automation.

History

History about this web site

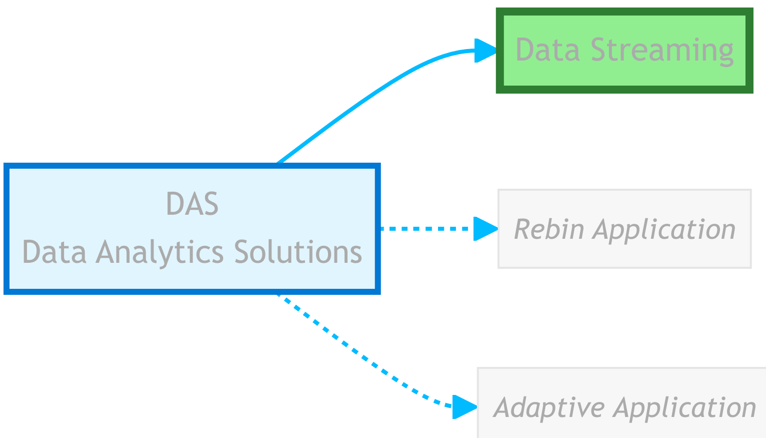


Data Streaming

Understand how the data stream delivers real-time access to comprehensive tester data.

> ⓘ **NOTE**
>
> **DAS Definition:** A DAS (Data Analytics Solution) is a software application that subscribes to Archimedes data streams to analyze tester data and drive insights or actions.

This chapter focuses on the data streaming capabilities within the DAS ecosystem, as illustrated below:



This section covers:

1. [DAS Overview - Understanding Data Analytic Solutions]
2. [Advanced DAS Concepts]
3. [Data Format and Structure]
4. [Implementing a DAS Listener]
5. [URL Reservation]

This documentation is part of the Teradyne Archimedes Reference Architecture series.

UltraEdge2000 Reference Architectures

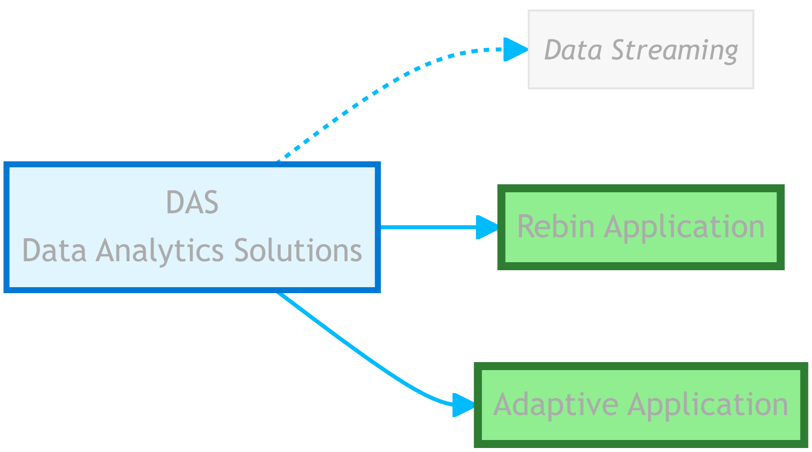
Explore UltraEdge architectures and the communication models used to integrate with UltraEdge

The Teradyne UltraEdge 2000 PC is a highly secure and powerful computer designed to bring edge-computing to a test cell. This is a stateless PC that executes applications in its RAM disk thus increasing security (once the session ends, nothing is kept on the machine).
Combined with AMP, the UltraEdge2000 PC enables uses to run machine learning algorithms in a fully IP-protected manner.

Check out the menu on the left to access the available RAs about the UltraEdge2000.

> ⓘ **NOTE**
>
> **DAS Definition:** A DAS (Data Analytics Solution) is a software application that subscribes to Archimedes data streams to analyze tester data and drive insights or actions.

This chapter focuses on the test program interactions within the DAS ecosystem, as illustrated below:



Test Program Interactions

Understand how the data stream delivers real-time access to comprehensive tester data.

Adaptive Test

Rebin Application

Explore the way to interact with your test program without modifying your test program.

Explore the way to change the binning on the fly of your devices

Use Cases

Real-World Use Cases

Browse practical implementations and scenarios that show how Archimedes enables faster insights and action.

🎬

Adaptive Test with UltraEdge

Watch a comprehensive video demonstration of how Adaptive Test can be implemented with UltraEdge technology to create intelligent, self-optimizing test programs.

🎬

REbin Application based on GDBN Algorithm

Watch a comprehensive video demonstration of how Rebin Application can be implemented with the GDBN technology.

AI/ML Overview

1. Concepts in Artificial Intelligence
2. Fundamentals of Machine Learning
3. AI/ML Glossary

This documentation is part of the Teradyne Archimedes Reference Architecture series.

Glossary

Welcome to the Glossary section. This contains comprehensive definitions of technical terms and acronyms used throughout the Archimedes Reference Architecture.

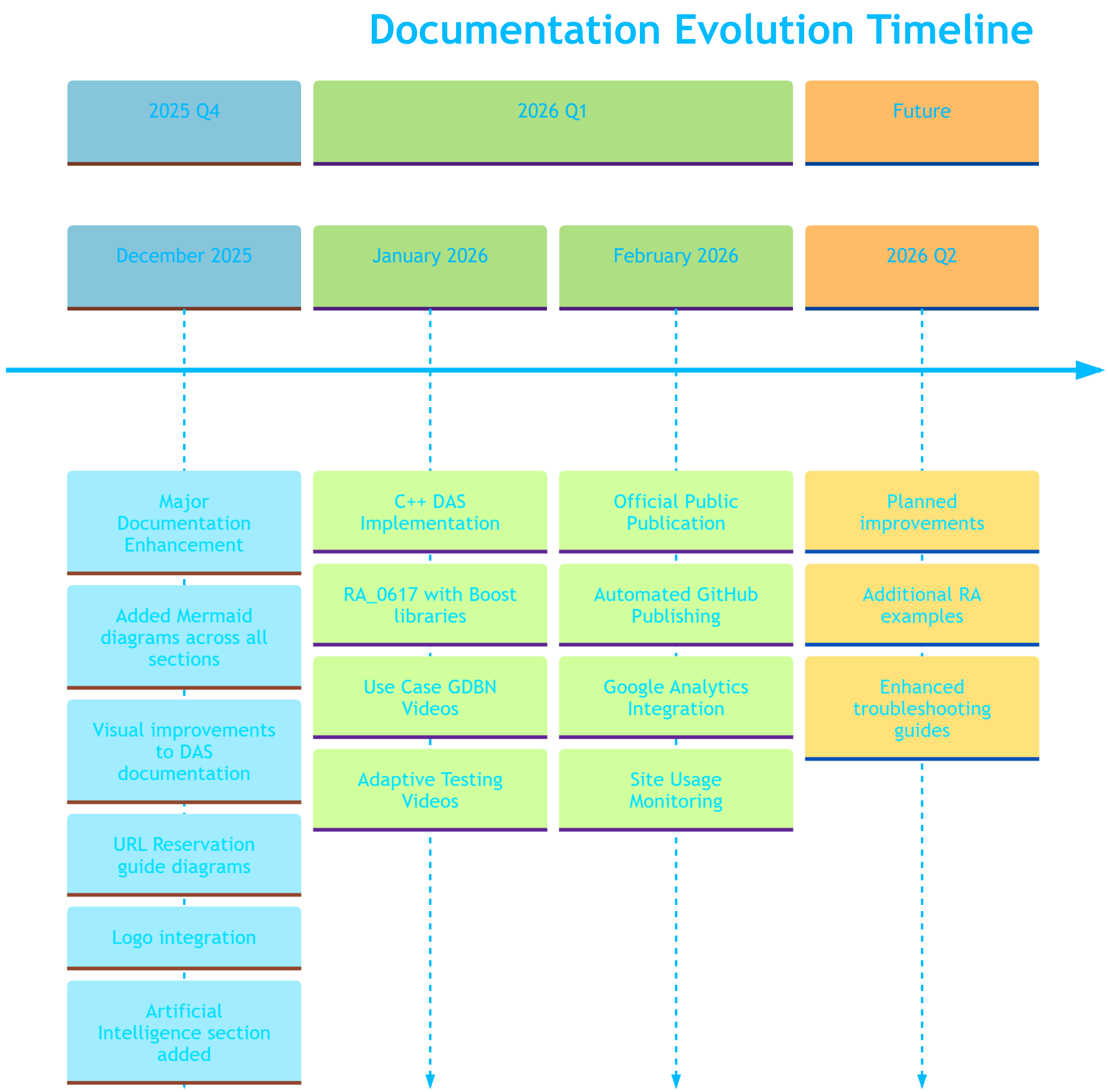
Available Glossaries

- Technical Glossary - Terms related to DAS, AMP, testing, and protocols

History

The Current Version of this web site is **1.1.1**

Timeline



What's New

May 2026 - C++ DAS Improvement 🚀

Review the RA 0617 related to C++ DAS

February 2026 - Official Public Publication 🚀

First official publication to public GitHub repository with analytics:

- ✅ **Public GitHub Publication**: Documentation now officially published to public repository
- ✅ **Automated Publishing**: Created scripts for building and publishing workflow
- ✅ **Google Analytics**: Integrated GA4 for monitoring site usage and audience insights
- ✅ **Usage Analytics**: Site visits and page views are now tracked to improve documentation quality

> **Note**: This documentation site uses Google Analytics to understand how users interact with the content, helping us improve documentation quality and relevance.

January 2026 - C++ DAS and Video Content

Expanded reference architectures and multimedia content:

- ✅ **C++ DAS Implementation**: Added RA_0617 - High-performance DAS using C++23 and Boost
- ✅ **Mermaid Diagrams**: Replaced ASCII diagrams with interactive Mermaid charts
- ✅ **Cross-Platform Build**: Zig build system for Windows/Linux compilation
- ✅ **GDBN Videos**: Video demonstrations for Good Die Bad Neighbor analysis
- ✅ **Adaptive Test Videos**: Tutorial videos showcasing adaptive testing workflows

December 2025 - Visual Documentation Enhancement

Febr A major effort to improve documentation clarity through visual diagrams:

- ✅ **DAS Communication**: Added sequence diagrams and architecture views
- ✅ **JSON Messages**: Created mindmaps and structure diagrams
- ✅ **DAS Listener Concept**: Organized messages with categorized mindmaps
- ✅ **URL Reservations**: Added decision trees and troubleshooting flowcharts
- ✅ **RA_0466**: Documented Python DAS implementation with diagrams
- ✅ **Branding**: Integrated Teradyne logo across all pages
- ✅ **Artificial Intelligence**: Added new AI & Machine Learning section to main navigation

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Last Updated: January 2026